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1 Relative Strength Index

Relative strength is the ratio of average upward price movement to average downward price movement over a selected period:

$$RS = \frac{\text{Average gain}}{\text{Average loss}}.$$

Suppose the relative strength is

$$RS = 2.0.$$

This means that over the measured period, upward price movement has been twice as large as downward price movement.

- $RS > 1$: gains have been larger than losses.
- $RS = 1$: gains and losses have been equal.
- $RS < 1$: losses have been larger than gains.
- $RS \rightarrow \infty$: there have been gains but essentially no losses.
- $RS = 0$: there have been losses but no gains.

The Relative Strength Index, or RSI, converts relative strength into a bounded value between 0 and 100:

$$RSI = 100 - \frac{100}{1 + RS} = 100 \left(\frac{RS}{1 + RS} \right).$$

As $RS \rightarrow \infty$, $RS/(1 + RS) \rightarrow 1$, so $RSI \rightarrow 100$. When $RS = 1$, $RSI = 50$. When $RS \rightarrow 0$, $RSI \rightarrow 0$.

Relative Strength	RSI
0	0
0.25	20
0.5	33.33
1	50
2	66.67
3	75
4	80
Very large	Approaches 100

RSI is commonly interpreted as follows:

- **Above 70:** frequently described as overbought.
- **Below 30:** frequently described as oversold.
- **Above 50:** upward momentum is dominant.
- **Below 50:** downward momentum is dominant.

To calculate the RSI for a time t , we need

$$RSI_t = \frac{\text{AvgGain}_t}{\text{AvgLoss}_t}.$$

For each candlestick c_j , compute the close-to-close change.

$$\Delta_{c_j} = \text{Close}_{c_j} - \text{Close}_{c_{j-1}}.$$

The change is called a gain if it is greater than zero, it is called a loss if it is less than zero.

$$\text{Gain}_{c_j} = \max(\Delta_{c_j}, 0), \quad \text{Loss}_{c_j} = \max(-\Delta_{c_j}, 0).$$

Then, compute (for a 14-period RSI)

$$\text{AvgGain}_{14} = \frac{\sum_{j=1}^{14} \text{Gain}_{c_j}}{14}, \quad \text{AvgLoss}_{14} = \frac{\sum_{j=1}^{14} \text{Loss}_{c_j}}{14}.$$

1.1 Wilder's smoothing

Wilder's smoothing is a recursive moving-average method used in RSI, ATR, and several other indicators. It gives more weight to recent observations while retaining information from all earlier observations.

For a period n , the update formula is

$$S_t = \frac{(n-1)S_{t-1} + X_t}{n}.$$

- X_t is the newest observation.
- S_{t-1} is the previous smoothed value.
- S_t is the new smoothed value.
- n is the smoothing period.

2 VWAP (VOLUME WEIGHTED AVERAGE PRICE)

For the relative strength, instead of using

$$\text{AvgGain}_{14} = \frac{\sum_{j=1}^{14} \text{Gain}_{c_j}}{14}, \quad \text{AvgLoss}_{14} = \frac{\sum_{j=1}^{14} \text{Loss}_{c_j}}{14},$$

we use Wilder's smoothing:

$$\begin{aligned} \text{AvgGain}_{c_j} &= \frac{13\text{AvgGain}_{c_{j-1}} + \text{Gain}_{c_j}}{14} \\ \text{AvgLoss}_{c_j} &= \frac{13\text{AvgLoss}_{c_{j-1}} + \text{Loss}_{c_j}}{14}. \end{aligned}$$

Then,

$$RS_{c_j} = \frac{\text{AvgGain}_{c_j}}{\text{AvgLoss}_{c_j}}.$$

Consequently,

$$RSI_{c_j} = 100 \left(\frac{RS_{c_j}}{1 + RS_{c_j}} \right).$$

The recursive formula requires a previous smoothed value, so the first value is normally initialized with a simple average of the first n observations:

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

For the RSI, the initial average gain and initial average loss are simply the averages

$$\text{AvgGain}_{14} = \frac{\sum_{j=1}^{14} \text{Gain}_{c_j}}{14}, \quad \text{AvgLoss}_{14} = \frac{\sum_{j=1}^{14} \text{Loss}_{c_j}}{14}.$$

Every later value uses Wilder's recursive update.

2 VWAP (Volume Weighted Average Price)

The average price a stock has traded at throughout the day weighted by volume. Instead of treating each price the same, VWAP assigns more weight to prices where trading activity was heavier.